

Last Time

Permutations and Combinations

Random Variables

Distributions

- Discrete Distributions

- Continuous Distributions

Relationships Between Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Today

Random Variables

Distributions

- Discrete Distributions

- Continuous Distributions

Relationships Between Distributions

QQ Plots

Random Variable

A random variable X is a mapping between a sample space S and the real numbers $\mathcal{R}$:

$$X : S \rightarrow \mathcal{R}.$$

Discrete Random Variables

For a discrete random variable X :

- ▶ The sample space S is finite or countably infinite.
- ▶ Probability function
 1. $0 \leq p_X(r) \leq 1.0$ for all $r \in \mathcal{R}$
 2. $\sum_{r \in S} p_X(r) = 1.0$
 3. $Pr(X = r) = p_X(r)$

Continuous Random Variables

For a continuous random variable X :

- ▶ The *support* S is uncountably infinite (range(s) of real numbers).
- ▶ Probability density function
 1. $f_X(r) \geq 0$ for all $r \in \mathcal{R}$
 2. $\int_{-\infty}^{\infty} f_X(r) dr = 1.0$
 3. $Pr(X = r) = 0$

Cumulative Distributions

$$P_X(r) = \sum_{x \leq r} p_X(x)$$

$$F_X(r) = \int_{-\infty}^r f_X(x) dx$$

Expected Value

The population mean μ

Discrete:

$$E(X) = \sum_{x \in S} xp_X(x)$$

Continuous:

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)$$

- ▶ $E(c) = c$
- ▶ $E(X + c) = E(X) + c$
- ▶ $E(cX) = cE(X)$

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- ▶ $E(X + c) = E(X) + c$
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$$E(aX + bY) = aE(X) + bE(Y)$$

Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[(X - E(X))^2]$$

- ▶ $\text{Var}(c) = 0$
- ▶ $\text{Var}(X + c) = \text{Var}(X)$
- ▶ $\text{Var}(cX) = c^2 \text{Var}(X)$

Variance

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If X and Y are independent,

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Moments

	Raw	Central	Related Statistic
1	$E(X)$	$E(X)$	Mean
2	$E(X^2)$	$E[(X - E(X))^2]$	Variance
3	$E(X^3)$	$E[(X - E(X))^3]$	Skewness
4	$E(X^4)$	$E[(X - E(X))^4]$	Kurtosis

Distributions

Discrete	Continuous
Uniform	Uniform
Bernoulli	Normal
Binomial	Student's t
Poisson	χ^2
Tabulated	F

Uniform

Roll a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$m = |S| = 6$$

$$p_X(x|m) = \frac{1}{m} \text{ for } x \in S, 0 \text{ else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b - a + 1} \text{ for } x \in S, \text{ 0 else}$$

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$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b-a+1} \text{ for } x \in S, \text{ 0 else}$$

$$E(X|a, b) = \frac{a+b}{2}$$

$$\text{Var}(X|a, b) = \frac{(b-a+1)^2 - 1}{12}$$

Bernoulli

A single experiment with only two possible outcomes: success and failure.
Flip a coin, call “heads” a success.

$$\begin{aligned} S &= \{0, 1\} \\ 0 &\leq p \leq 1 \\ p_X(x|p) &= \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= p \\ \text{Var}(X) &= p(1 - p) \end{aligned}$$

Binomial

Add up a sequence of N independent Bernoulli (0 or 1) experiments: flip a coin N times, answer N true/false questions. How many “successes” were there in the sequence?

$$S = \{0, 1, \dots, N\}$$

$$0 \leq p \leq 1$$

$$N \geq 1, N \in \mathcal{N}^+$$

$$p_X(x|N, p) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x} & x \in S \\ 0 & \text{otherwise.} \end{cases}$$

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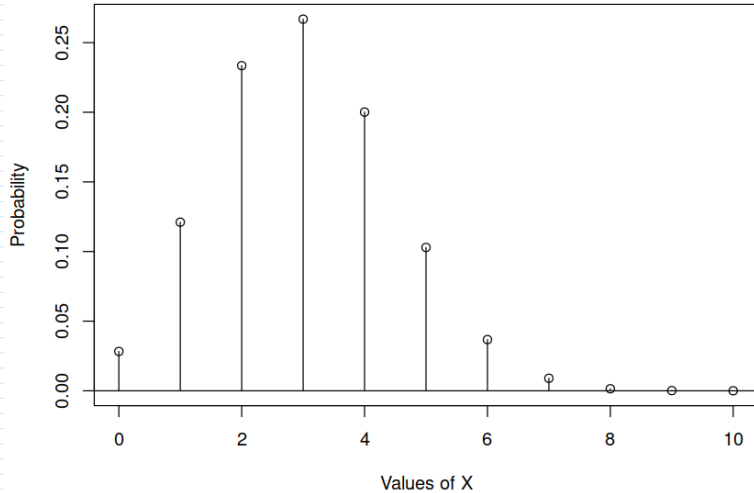
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$$E(X) = Np$$

$$\text{Var}(X) = Np(1-p)$$

Binomial



Poisson

- ▶ The number of rare events occurring in a fixed interval of time or bounded region of space. Airline accidents, hospital admissions, school shootings, typos on a web page, defective widgets in a shipment, etc.
- ▶ The limiting distribution of the binomial when $N \rightarrow \infty$ and $p \rightarrow 0$.

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$$S = \{0, 1, 2, \dots\}$$

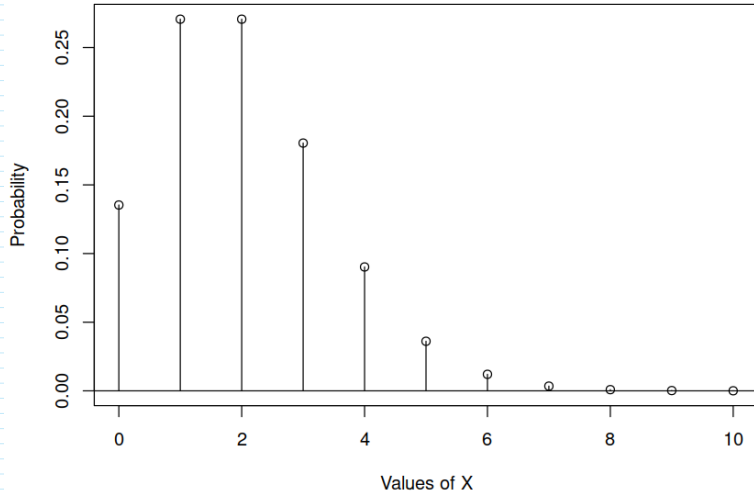
$$\lambda > 0$$

$$p_X(x|\lambda) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x \in S \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

Poisson



Tabled

A random variable that can take on values in a list, with each value x associated with a probability π_x . Relative frequency distributions are examples of tabled probability distributions.

GSS Health Rating Responses

x	1	2	3	4
π_x	0.37	0.46	0.15	0.02

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$$S = \{1, 2, 3, 4\}$$

$$p_X(x|\pi_1, \pi_2, \pi_3, \pi_4) = \begin{cases} \pi_1 & x = 1 \\ \vdots & \\ \pi_4 & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \sum_{x=1}^N x\pi_x$$

$$Var(X) = \sum_{x=1}^N (x - E(X))^2 \pi_x$$

Uniform

- ▶ A “rectangular” distribution with support equal to a range; all values in the range have equal density.
- ▶ Important for statistical applications: generation of random numbers, Bayesian priors

$$\begin{aligned} S &= [a, b] \\ a &< b \\ f_X(x|a, b) &= \begin{cases} \frac{1}{b-a} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= \frac{a+b}{2} \\ \text{Var}(X) &= \frac{(b-a)^2}{12} \end{aligned}$$

Normal

- ▶ The most important distribution for us!
- ▶ Many natural systems result in normally distributed variables.
- ▶ Central Limit Theorem

$$S = (-\infty, \infty) = \mathcal{R}$$

$$\mu \in \mathcal{R}$$

$$\sigma^2 > 0$$

$$f_X(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for } x \in S$$

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$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

Standard Normal

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\text{Z-score of } X: Z_X = \frac{X - \mu}{\sigma}$$

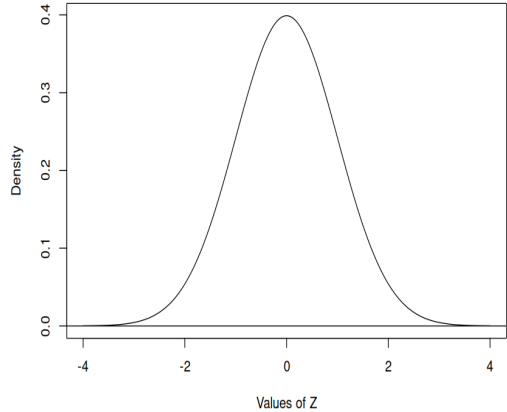
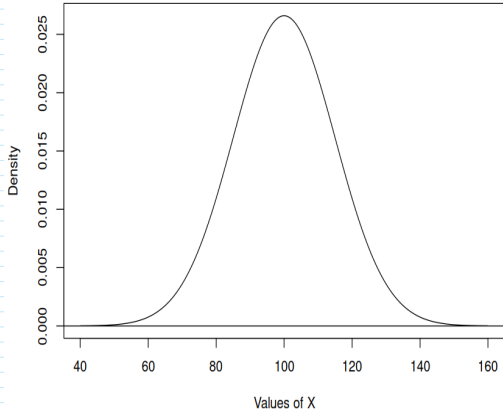
$$S = (-\infty, \infty) = \mathcal{R}$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \text{ for } z \in S$$

$$E(Z) = 0$$

$$\text{Var}(Z) = 1$$

Normal and Standard Normal



Student's t

- ▶ William Sealy Gosset (1876-1937)
- ▶ A bell-shaped distribution used when the population variance σ^2 is estimated by the statistic s^2 .

$$\begin{aligned} S &= (-\infty, \infty) = \mathcal{R} \\ \nu &\in \{0, 1, 2, \dots\} \\ f_X(x|\nu) &= \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad \text{for } x \in S \end{aligned}$$

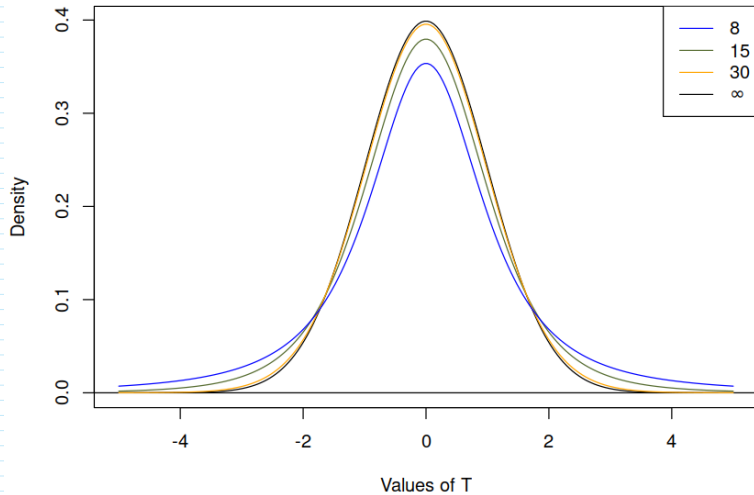
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$$\begin{aligned} E(X) &= \begin{cases} 0 & \nu > 1 \\ \text{undefined} & \text{otherwise} \end{cases} \\ \text{Var}(X) &= \begin{cases} \frac{\nu}{\nu-2} & \nu > 2 \\ \infty & 1 < \nu \leq 2 \\ \text{undefined} & \text{otherwise} \end{cases} \end{aligned}$$

Student's t



χ^2

The distribution of a sum of squared z -scores, related to the distribution of the sample variance.

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ k &\in \{1, 2, \dots\} \\ f_X(x|k) &= \begin{cases} \frac{x^{k/2-1}}{2^{k/2}\Gamma(k/2)} e^{-x/2} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

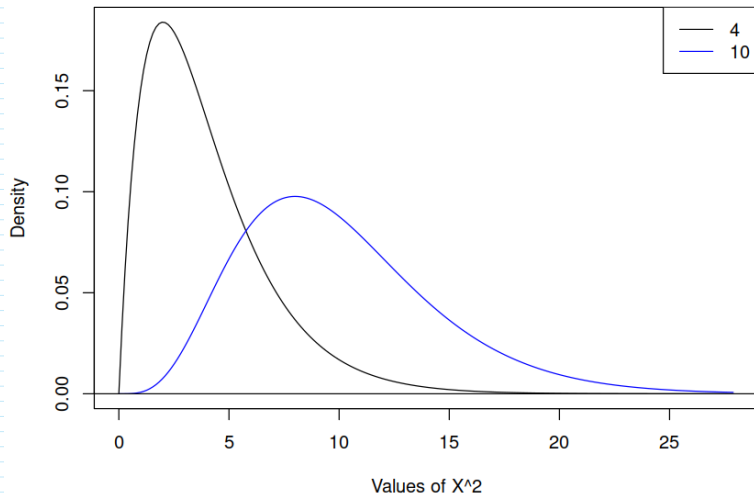
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$$\begin{aligned} E(X) &= k \\ \text{Var}(X) &= 2k \end{aligned}$$

χ^2



F

- ▶ A distribution related to the ratio of two sample variances,
- ▶ The basis for ANOVA, regression model comparison, etc

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ \nu, \delta &> 0 \\ f_X(x|\nu, \delta) &= \begin{cases} \frac{\Gamma\left(\frac{\nu + \delta}{2}\right) \nu^{\nu/2} \delta^{\delta/2}}{\Gamma\left(\frac{\nu}{2}\right) \Gamma\left(\frac{\delta}{2}\right)} \frac{x^{\nu/2-1}}{(\delta + \nu x)^{(\nu+\delta)/2}} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

F

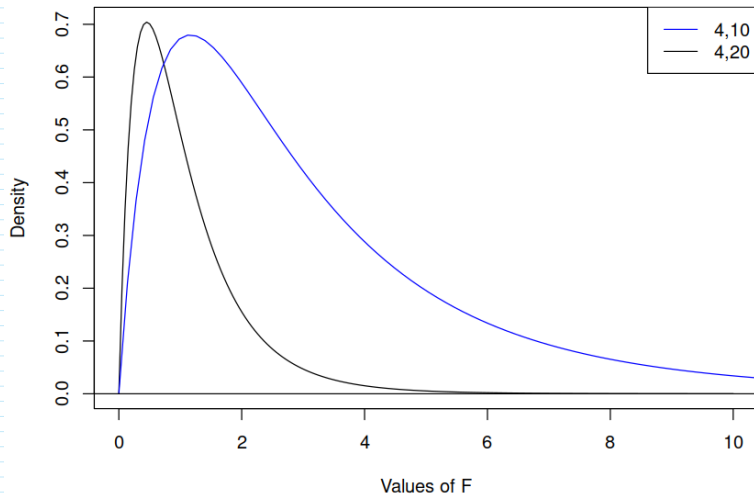
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$$\begin{aligned} E(X) &= \frac{\delta}{\delta - 2} \text{ for } \delta > 2 \\ \text{Var}(X) &= \frac{2\delta^2(\nu + \delta - 2)}{\nu(\delta - 2)^2(\delta - 4)} \text{ for } \delta > 4 \end{aligned}$$

If $\delta \leq 2$ the mean does not exist. If $\delta \leq 4$ the variance doesn't exist.

F



Notation

Discrete	Continuous
Uniform $U(a, b)$	Uniform $U(a, b)$
Bernoulli $Ber(p)$	Normal $\mathcal{N}(\mu, \sigma^2)$
Binomial $Bin(N, p)$	t $t(\nu)$
Poisson $Pois(\lambda)$	χ^2 $\chi^2(k)$
Tabulated $p_X(x p_1, \dots, p_m)$	F $F(\nu, \delta)$

Relationships Among Distributions

Leemis Chart

- ▶ $X \sim \text{Ber}(p): \sum_{i=1}^N X_i \sim \text{Bin}(N, p)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty, Np = \lambda} \text{Bin}(N, p) \sim \text{Pois}(\lambda)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty} \text{Bin}(N, p) \sim \mathcal{N}(Np, Np(1 - p))$
- ▶ $Z \sim \mathcal{N}(0, 1): \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$
- ▶ $X \sim t(\nu): X^2 \sim F(1, \nu)$
- ▶ $X \sim t(\nu): \lim_{\nu \rightarrow \infty} t(\nu) \sim \mathcal{N}(0, 1)$
- ▶ $X^2 \sim \chi^2(k): \lim_{k \rightarrow \infty} \chi^2(k) \sim \mathcal{N}(k, 2k)$
- ▶ $X_1^2 \sim \chi^2(n_1), X_2^2 \sim \chi^2(n_2): \frac{X_1^2/n_1}{X_2^2/n_2} \sim F(n_1, n_2)$

QQ Plot

- ▶ Plot the quantiles of the observed data (y axis) against the quantiles of the theoretical distribution (x axis).
- ▶ The plot points should fall on the identity line $y = x$.

R code

```
data <- read.csv("GSS_1980.csv",header=TRUE)
age <- data$AGE

# Pick some good values for the gamma distribution
a <- mean(age)/var(age)
k <- mean(age)^2/var(age)

# Compute the quantiles for the data and the gamma model
obs <- quantile(age,seq(.025,.975,by=.05))
theory <- qgamma(seq(.025,.975,by=.05),k,a)

# Plot the data histogram with the gamma model
par(mfrow=c(1,2))
hist(age,freq=FALSE,breaks=seq(15.5,90.5,by=5),
      xlab="age",ylab="Density",main="")
lines(0:80,dgamma(0:80,k,a))

# QQ Plot
plot(theory,obs,type="p",ylab="Age Quantiles",
      xlab="Theoretical Gamma Quantiles")
abline(0,1)
```

QQ Plot

